Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science Year 10 2026

The Kitchen CPU

A Human CPU play script · Run 1

LOAD 5, ADD 6, STORE 7, STOP

Characters

Technical name	Short	Character	Job
Control Unit	CU	**Head Chef**	Decodes each step with the codebook and gives the orders. Never touches the food.
Program Counter	PC	**Line Counter**	Knows which line comes next. Adds 1 after every fetch.
Memory Address Register	MAR	**Caller**	Holds the line number and calls it out to the recipe card.
Memory Data Register	MDR	**Scribe**	Copies down whatever comes back from the card: an instruction or a number.
Main memory	RAM	**Librarian**	Keeps the recipe card, outside the CPU. A line's number is its address. Reading a line leaves it there.
Arithmetic Logic Unit	ALU	**Station Cook**	Does the sums, like 12 + 30.
Accumulator	ACC	**Bowl**	Sits on the scales. The reading is the running total.

The recipe card *(main memory)*

Line	Shorthand	The Head Chef reads it as
0	`LOAD 5`	Put the amount on line 5 into the bowl
1	`ADD 6`	Add the amount on line 6 to the bowl
2	`STORE 7`	Write the bowl's total on line 7
3	`STOP`	The recipe is finished
4		*blank*
5	`12`	12 g
6	`30`	30 g
7		Total: _____

The codebook *(instruction set)*

`LOAD n`	Put the amount on line n into the bowl.
`ADD n`	Add the amount on line n to the bowl.
`STORE n`	Write the bowl's total on line n.
`STOP`	The recipe is finished.

Only the Head Chef holds the codebook. Anyone can carry or read out a line; only the Head Chef can turn it into orders.

Step	Part	Character	Says or does

Scene 1: LOAD 5

Lights up on a busy kitchen. The Head Chef stands at the pass. The Librarian stands across the room, holding the recipe card. The Line Counter, the Caller and the Scribe wait by the pass. The Station Cook and the Bowl stand at the scales.

The Line Counter says 0. The Caller says empty. The Scribe says empty. The Bowl says empty.

FETCH *get it*

F1	**CU**	**Head Chef:**	*(nothing to cook)* OK, team, let's get started. Line Counter, call out our line!
F2	**PC**	**Line Counter:**	*(to the Caller)* We're on line zero!
F3	**MAR**	**Caller:**	*(wipes the board and writes 0)* Got it! Librarian, please prep line zero for **reading**. Await your command.
F4	**CU**	**Head Chef:**	*(to the Librarian)* Approved! **Read!**
F5	**RAM**	**Librarian:**	*(reads the line out and keeps the card)* LOAD 5.
F6	**MDR**	**Scribe:**	*(writes LOAD 5 down on the notepad)*
F7	**PC**	**Line Counter:**	*(in the meantime, adds 1 and writes 1, ready for the next call)*

DECODE *understand it*

D1	**MDR**	**Scribe:**	*(holds up the notepad for the Head Chef)* LOAD 5.
D2	**CU**	**Head Chef:**	*(checks the codebook)* LOAD 5 means: put the amount on line five into the bowl.

EXECUTE *do it*

E1	**CU**	**Head Chef:**	Caller, I need line 5.
E2	**MAR**	**Caller:**	*(wipes the board and writes 5)* Got it! Librarian, please prep line 5 for reading. Await your command.
E3	**CU**	**Head Chef:**	*(to the Librarian)* **Read!**
E4	**RAM**	**Librarian:**	*(reads the line out and keeps the card)* 12 grams.
E5	**MDR**	**Scribe:**	*(writes 12 grams on the notepad)*
E6	**CU**	**Head Chef:**	*(to the Bowl)* Bowl, you are asked to hold 12 grams.
E7	**ACC**	**Bowl:**	12 grams! Got it! *(12 g is shown on the scales.)*

Scene 2: ADD 6

The Line Counter says 1. The Caller says 5. The Scribe says 12. The Bowl says 12.

FETCH *get it*

F1	**CU**	**Head Chef:**	OK, team, let's continue. Line Counter, call out our line!
F2	**PC**	**Line Counter:**	*(to the Caller)* We're on line one!
F3	**MAR**	**Caller:**	*(wipes the board and writes 1)* Got it! Librarian, please prep line one for **reading**. Await your command.
F4	**CU**	**Head Chef:**	*(to the Librarian)* Approved! **Read!**
F5	**RAM**	**Librarian:**	*(reads the line out and keeps the card)* ADD 6.
F6	**MDR**	**Scribe:**	*(writes ADD 6 down on the notepad)*
F7	**PC**	**Line Counter:**	*(in the meantime, adds 1 and writes 2, ready for the next call)*

DECODE *understand it*

D1	**MDR**	**Scribe:**	*(holds up the notepad for the Head Chef)* ADD 6.
D2	**CU**	**Head Chef:**	*(checks the codebook)* ADD 6 means: add the amount on line six to the bowl.

EXECUTE *do it*

E1	**CU**	**Head Chef:**	Caller, I need line 6.
E2	**MAR**	**Caller:**	*(wipes the board and writes 6)* Got it! Librarian, please prep line 6 for reading. Await your command.
E3	**CU**	**Head Chef:**	*(to the Librarian)* **Read!**
E4	**RAM**	**Librarian:**	*(reads the line out and keeps the card)* 30 grams.
E5	**MDR**	**Scribe:**	*(writes 30 grams on the notepad)*
E6	**CU**	**Head Chef:**	*(to the Station Cook)* Station Cook, take what is in the bowl now and add the next ingredients. How much is that?
E7	**ALU**	**Station Cook:**	*(reads the Bowl and the Scribe's notepad)* We had 12 grams in the bowl already, sir, and when we add 30 more grams it adds up to **42 grams**. Bowl, you now have 42 grams!
E8	**ACC**	**Bowl:**	42 grams! Got it! *(42 g is shown on the scales.)*

Scene 3: STORE 7

The Line Counter says 2. The Caller says 6. The Scribe says 30. The Bowl says 42.

FETCH *get it*

F1 **CU** **Head Chef:** OK, team, let's continue. Line Counter, call out our line!

F2 **PC** **Line Counter:** *(to the Caller)* We're on line two!

F3 **MAR** **Caller:** *(wipes the board and writes 2)* Got it! Librarian, please prep line two for **reading**. Await your command.

F4 **CU** **Head Chef:** *(to the Librarian)* Approved! **Read!**

F5 **RAM** **Librarian:** *(reads the line out and keeps the card)* STORE 7.

F6 **MDR** **Scribe:** *(writes STORE 7 down on the notepad)*

F7 **PC** **Line Counter:** *(in the meantime, adds 1 and writes 3, ready for the next call)*

DECODE *understand it*

D1 **MDR** **Scribe:** *(holds up the notepad for the Head Chef)* STORE 7.

D2 **CU** **Head Chef:** *(checks the codebook)* STORE 7 means: **write** the bowl's total on line seven.

EXECUTE *do it*

E1 **CU** **Head Chef:** Caller, please prepare the Librarian for line 7.

E2 **MAR** **Caller:** *(wipes the board and writes 7)* Librarian, please prep line 7 for **writing**. Await your command.

No command is given by the Head Chef yet. Librarian is waiting.

E3 **CU** **Head Chef:** Bowl, show the Scribe your total.

E4 **ACC** **Bowl:** *(shows 42 to the Scribe and keeps its number)*

E5 **MDR** **Scribe:** *(copies 42 onto the notepad)*

E6 **MDR** **Scribe:** *(walks over and hands the number to the Librarian)* Forty-two.

E7 **CU** **Head Chef:** *(to the Librarian)* **Write!**

E8 **RAM** **Librarian:** *(writes 42 in line 7's box, in pencil)*

Scene 4: STOP

The Line Counter says 3. The Caller says 7. The Scribe says 42. The Bowl says 42.

FETCH *get it*

F1	**CU**	**Head Chef:**	OK, team, let's continue. Line Counter, call out our line!
F2	**PC**	**Line Counter:**	*(to the Caller)* We're on line three!
F3	**MAR**	**Caller:**	*(wipes the board and writes 3)* Got it! Librarian, please prep line three for **reading**. Await your command.
F4	**CU**	**Head Chef:**	*(to the Librarian)* Approved! **Read!**
F5	**RAM**	**Librarian:**	*(reads the line out and keeps the card)* STOP.
F6	**MDR**	**Scribe:**	*(writes STOP down on the notepad)*
F7	**PC**	**Line Counter:**	*(in the meantime, adds 1 and writes 4, ready for the next call)*

DECODE *understand it*

D1	**MDR**	**Scribe:**	*(holds up the notepad for the Head Chef)* STOP.
D2	**CU**	**Head Chef:**	*(checks the codebook)* STOP means: the recipe is finished.

EXECUTE *do it*

E1	**CU**	**Head Chef:**	*(closes the codebook)* OK, everyone, we are done! The Librarian says to stop, so we stop. Well done, team!

Lights down. Line 7 now reads 42.